

Training-Induced Occipital Recruitment in Late-Blind Adults

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Cross modal plasticity permits the visual cortex to process non visual input after blindness, yet the degree to which adults who lose vision late in life can reengage occipital regions remains unresolved. I propose an eight-week multisensory navigation training that pairs real-time auditory distance tones with vibrotactile directional cues. Thirty late blind adults and matched blindfolded sighted controls will complete twenty-four one-hour sessions. Functional MRI and behavioural navigation tests will be acquired pre and post-training. We predict increased V1/V2 blood-oxygen-level-dependent activity and stronger occipito-auditory connectivity in late-blind trainees, no such change in wait-list controls, and superior gains in real-world route efficiency for trainees versus controls. Demonstrating trainable occipital recruitment in late blindness would refine theories of sensitive periods and inform sensory-substitution rehabilitation, supporting policies that fund technology and training aimed at harnessing residual cortical plasticity.

The human brain exhibits a remarkable capacity to repurpose unused circuitry after sensory loss, cross-modal plasticity. In blindness, occipital regions traditionally dedicated to visual input can become engaged by auditory, tactile, or linguistic information. The extent of such reorganisation appears strongly influenced by the age at which vision is lost. Early-blind individuals, who lack visual input during critical periods, routinely recruit primary and extrastriate visual cortices when reading Braille or generating verbs. Late-blind adults often display more modest occipital involvement, suggesting that sensitive periods constrain large-scale takeover.

Sensitive-Period versus Residual Plasticity Accounts

fMRI shows that Braille reading robustly activates V1 in early but not late-blind adults (Sadato et al., 2002). Structural MRI further reveals a thicker calcarine cortex in adults blind since infancy, likely reflecting retained exuberant connections and reduced pruning (Jiang et al., 2009).

Connectivity analyses indicate that early blindness fosters occipital specialisation for auditory spatial processing, whereas late blindness preserves sight-like patterns (Collignon et al., 2013).

Together these findings support a sensitive-period model in which early deprivation permits extensive reorganisation, while deprivation after adolescence yields attenuated change.

A strict critical-period view, however, underestimates the latent plasticity of the adult visual cortex. Reviews of sensory-substitution research conclude that visual areas in late-blind individuals can still be reshaped by intensive training (Chan, 2018). Empirical studies now show functional and structural modifications in late-blind occipital cortex following targeted intervention, motivating a residual-plasticity account in which substantial change remains possible well into adulthood.

Training-Induced Plasticity in Late Blindness

Auditory echolocation. Norman et al. (2024) enrolled late-blind and sighted adults in a ten-week click-based echolocation course. Post-training scans showed enhanced V1 activation to echoes, paralleled by improved obstacle detection. Earlier case reports likewise documented strong calcarine responses during echolocation in a late-blind expert (Thaler et al., 2011).

Spatial sound localisation. Tao et al. (2017) trained late- and early-blind adults to localise broadband noises. Late-blind learners exhibited increased precuneus activation and stronger precuneus-to-lingual-gyrus connectivity after training, implying recruitment of visuospatial memory networks established during sighted years.

Multisensory sensory substitution. Chebat et al. (2015) taught blind participants—including many late blind—to navigate real and virtual mazes with the EyeCane, which converts distance to vibrotactile and auditory cues. Within three days, navigation errors fell to sighted-control levels. Complementing these behavioural gains, Ortiz et al. (2011) showed that three months of tactile pattern training evoked occipital EEG response and vivid "visual" sensations—in some late-blind adults, directly linking occipital reactivation to new perceptual experience.

Additional navigation interventions. Emerging work with tongue-display (BrainPort) and speech-encoded spatial sonification (Topo-Speech) systems reports that late-blind users can achieve above-80 % spatial-accuracy after weeks of practice, often accompanied by occipital BOLD increases. Collectively, these studies confirm that the visual cortex of late-blind adults is not dormant and can be re-engaged through sustained, task-relevant practice.

Building on this literature, we propose an eight-week multisensory navigation programme designed to strengthen occipital recruitment in late-blind adults. Participants will train with an auditory–tactile interface that supplies real-time distance and direction cues, enabling exploration of unfamiliar indoor routes. Occipital activation and connectivity will be assessed with functional MRI alongside real-world navigation performance. A sighted blindfolded group completes identical training to control nonspecific practice effects.

Functional activation – Post-training, late-blind participants will show increased BOLD in V1/V2 during an auditory spatial-navigation task, approaching early-blind activation levels.

Connectivity – Resting-state and task-based functional connectivity between occipital cortex and auditory/posterior-parietal areas will strengthen in the late-blind group.

Behaviour – Late-blind participants will display greater gains in navigation accuracy and speed than blindfolded sighted controls.

A power analysis in GPower $d = 0.80$, $\alpha = .05$, $1 - \beta = .80$, two-tailed, repeated-measures interaction, indicated a minimum of 26 participants. To allow attrition we will recruit thirty adults who became totally blind after age 15 ($M = 30$ years, $SD = 5$). Inclusion criteria- no light perception, normal hearing, no neurological or psychiatric diagnoses, and MRI compatibility. Participants are randomly assigned to Training ($n = 15$) or Wait-list Control ($n = 15$). A matched Sighted-Blindfold group ($n = 15$) controls generic learning effects. Baylor IRB approval will be taken

Apparatus and Sensory-Substitution Interface

The navigation aid combines a stereo bone-conduction headset distance-modulated tones 0.5–4, with a wristband housing three vibrotactile motors signalling obstacle direction (left, centre, right). A chest-mounted depth camera streams data to a microcontroller that updates cues at 30 Hz. The system weighs < 400 g and leaves hands free for cane use.

Training Protocol

Participants in Training and Sighted-Blindfold groups complete twenty-four one-hour sessions across eight weeks:

Exploration phase (20 min) – independent traversal of a novel 50 m indoor route.

Feedback phase (20 min) – instructor-guided repeats of error segments, emphasising cue interpretation.

Free-navigation phase (20 min) – timed point-to-point travel with verbal feedback.

The Wait-list Control group receives no training during this interval but may enrol later.

A schematic timeline illustrates assessment and training weeks.

Tasks and Behavioural Measures

Real-world navigation (Figure 2) is assessed pre-/post-training on three unseen routes; metrics: 1, completion time, 2 collisions, 3 wrong turns.

Virtual auditory navigation inside the MRI presents 3-D binaural audio scenes; participants move via a three-button box to locate a hidden beacon, permitting search-path-efficiency calculation.

MRI Acquisition and Pre-processing

Scanning is on a 3 T Siemens Prisma. Functional runs: gradient-echo EPI, TR = 2000 ms, TE = 30 ms, flip = 75°, 38 slices, 3 mm isotropic. High-resolution T1 (1 mm³) supports normalisation.

Pre-processing in SPM12: slice-timing, motion realignment, coregistration, MNI normalisation, 6 mm FWHM smoothing; ICA-AROMA removes motion artefacts.

Statistical Analysis

Functional activation: mixed 2 (Group $\times$ Time) ANOVA on mean V1/V2 percent-signal change.

Functional connectivity: psychophysiological interaction (PPI) analyses seeding V1; Fisher-z coefficients enter the same ANOVA.

Behaviour: 3 (Group) $\times$ 2 (Time) mixed ANOVAs on each navigation metric. Alpha = .05, Bonferroni-corrected.

Expected Results

Functional Activation

Figure 3A will plot mean V1/V2 percent-signal change (bars $\pm$ SE) showing a significant Group $\times$ Time interaction: late-blind trainees increase post-training, wait-list controls unchanged, sighted controls modest rise.

Functional Connectivity

Figure 3B will illustrate V1–A1 connectivity (line graph) revealing strengthened coupling only in late-blind trainees.

Behavioural Performance

Figure 3C will graph completion time, collisions, and wrong turns. Late-blind trainees reduce time $\approx 25\%$, halve collisions, and cut wrong turns relative to both control groups.

Discussion

This study examines whether eight weeks of multisensory navigation training can unmask visual-cortex plasticity in adults who lost sight after childhood. We predict enhanced occipital activation, strengthened occipito-auditory connectivity, and substantial behavioural gains, narrowing the gap with those blind since birth.

Integration With Prior Literature

Our hypotheses mirror training-induced V1 engagement in late-blind echolocation learners (Norman et al., 2024) and connectivity gains observed by Tao et al. (2017). Anticipated navigation improvements extend EyeCane findings (Chebat et al., 2015) to a longer regimen. Confirmation would strengthen residual-plasticity accounts by showing that adult visual cortex remains adaptable when given spatially rich input.

Theoretical Implications

Positive results would challenge rigid critical-period models, favouring a graded framework in which early deprivation enables maximal plasticity yet substantial reorganisation persists later, especially when training leverages top-down imagery networks established during sight.

Clinical, Technological, and Policy Significance

Trainable occipital recruitment could reshape rehabilitation for millions who lose vision after adolescence. Demonstrating efficacy may justify insurance coverage for sensory-substitution devices and funding for community-based training programmes, aligning policy with neuroscience-informed best practice.

Limitations and Future Directions

Sample size is modest; future work should include larger cohorts and longer-term follow-up. Diffusion-tensor imaging could test whether white-matter changes parallel functional gains. Comparing devices like the EyeCane, BrainPort, Topo-Speech may reveal which stimulus parameters optimise occipital plasticity.

Conclusion

If confirmed, our findings will demonstrate that structured multisensory training meaningfully re-engages visual cortex in late-blind adults, yielding real-world navigation benefits and informing theories of cortical adaptability.

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